Lierda NT26-FCN B series (external boost) Hardware Reference Design Manual

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Date: 25/08/07

Status: Released

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Revision History

Document version	Change date	Reviser	Reviewer	Change content
Rev1.0	25-08-07			Initial version,providing VBAT boost circuit.

Safety Instructions

Users are responsible for complying with the relevant regulations on wireless communication modules and devices in other countries and specific environmental regulations. By following the following safety principles, personal safety can be ensured and help protect products and work environments from potential damage. Our company is not responsible for any losses related to customers' failure to comply with these regulations.

	Road safety comes first! When driving, please do not use handheld mobile devices unless they have hands-free functionality. Please pull over before making a call!
	Please turn off your mobile terminal device before boarding. The wireless function of the mobile terminal is prohibited from being turned on in the airplane to prevent interference with the aircraft communication system. Ignoring this prompt may jeopardize flight safety, and even violate the law.
	When in a hospital or health care facility, pay attention to any restrictions on the use of mobile terminal devices. RF interference can cause medical equipment to malfunction, so it may be necessary to turn off mobile terminal devices.
	Mobile terminal devices do not guarantee effective connection in all situations, such as when there is no balance on the mobile terminal device or the SIM card is invalid. When you encounter the above situations in an emergency, remember to use emergency calls, and ensure that your device is turned on and in an area with sufficient signal strength.
	Your mobile terminal device will receive and emit radio frequency signals when it is powered on. When it is close to a TV, radio, computer, or other electronic devices, it will generate radio frequency interference.
	Please keep mobile terminal devices away from flammable gases. When you are near gas stations, oil depots, chemical plants, or explosive operation sites, please turn off the mobile terminal device. There is a safety hazard in operating electronic devices in any place with potential explosion risks.

Module selection for application

Serial number	Module model	Product model	Support frequency band	Dimensions (mm)	Module introduction
1	NT26-FCN	NT26FCNB10WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power off
2	NT26-FCN	NT26FCNB30WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power off Support VoLTE
3	NT26-FCN	NT26FCNB00WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power off
4	NT26-FCN	NT26FCNB70WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power off Support VoLTE
5	NT26-FCN	NT26FCNB60WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power off

Contents

1 Introduction

This document provides the application reference circuit for Lierda NT26-FCN B series Cat.1 module, as well as some notes on circuit design.

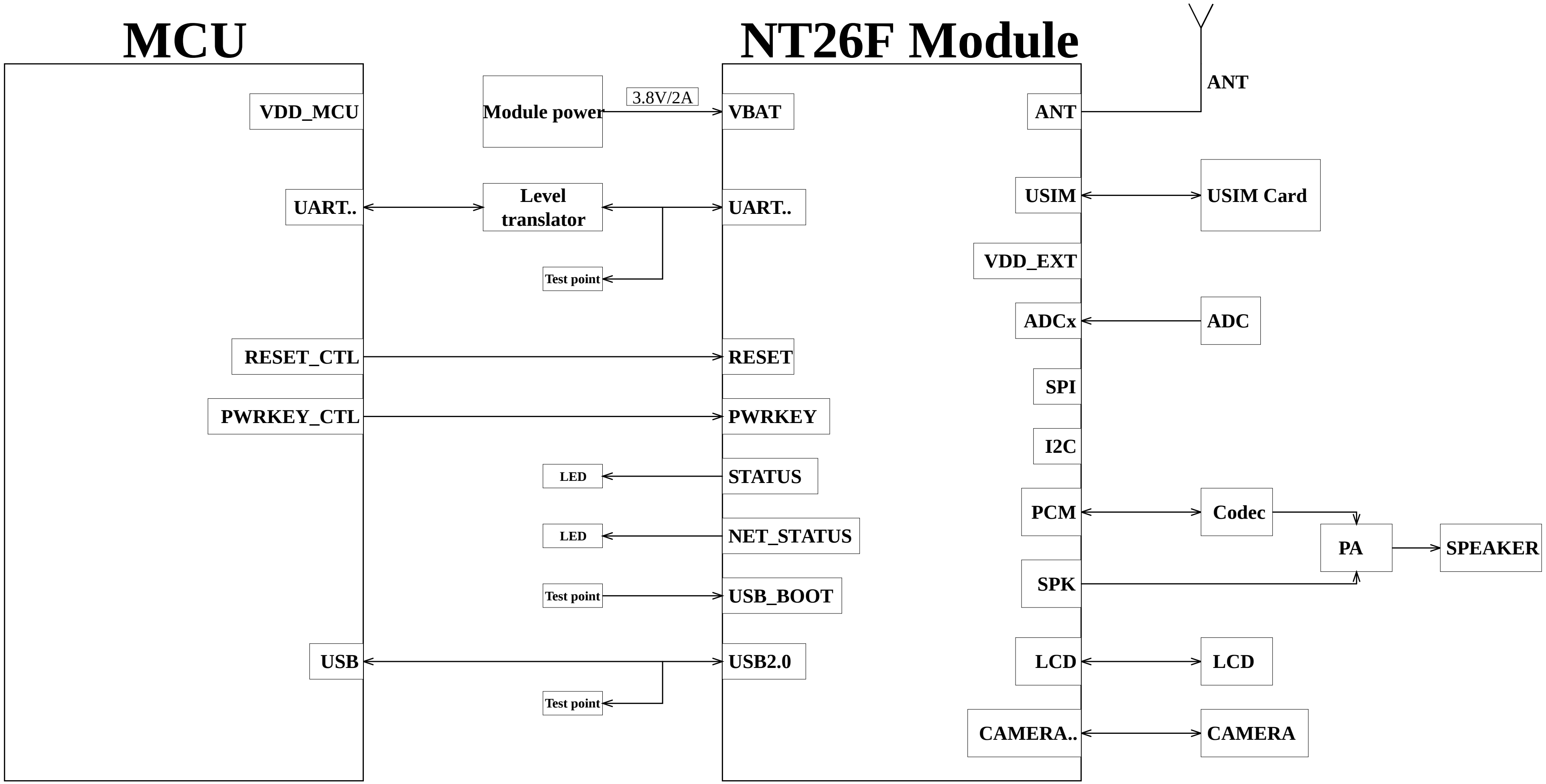
This document can help users quickly understand the peripheral hardware circuit of the module. Combined with other relevant documents, users can quickly master the application method of the Cat.1 module.

2 Reference design

This chapter is the hardware reference design of the NT26-FCN B series, mainly including power supply, SIM card, serial port, control signals, and peripheral application interface design.

2.1 Schematic diagram

The following is the reference design schematic of the NT26-FCN B series module, this design is for reference only.



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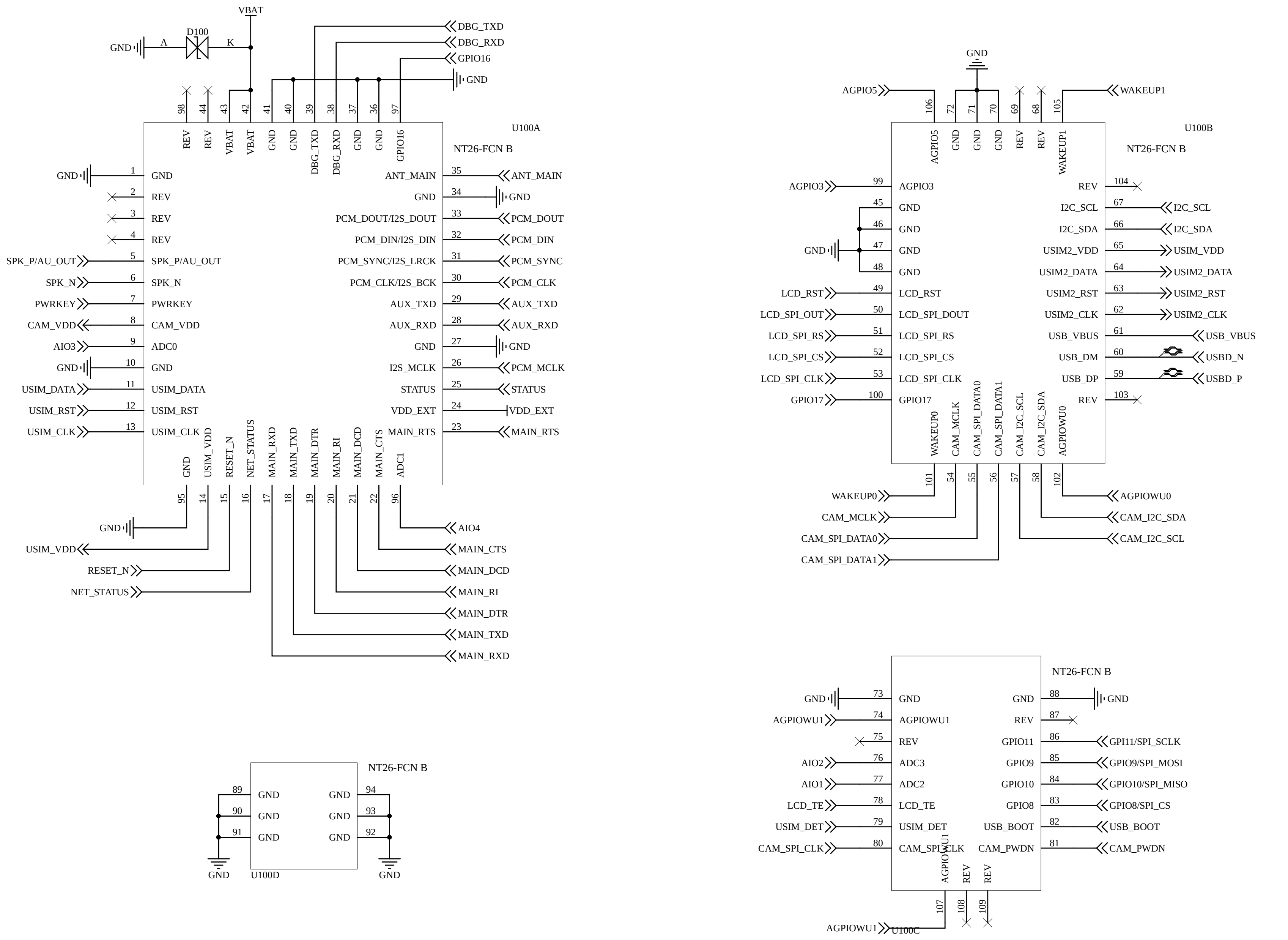

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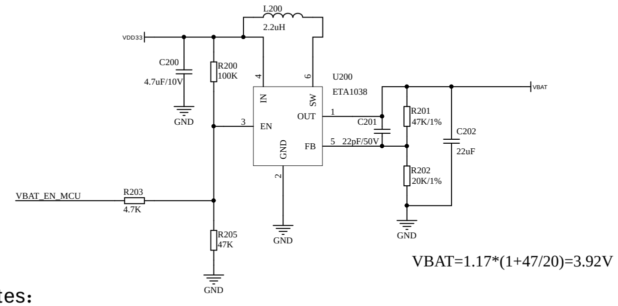
DC-DC BOOST

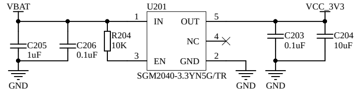


VBAT=1.17*(1+47/20)=3.92V

Notes:

(1) When the input power is lower than the working voltage of the module, select the boost type DC-DC power supply.
(2) This circuit can handle a maximum input voltage of 2.2-4.5V and the output current can reach 1.5A/5V.
(3) Pay attention to the selection of the L200 inductor. The saturation current should be set to above 3.5A.
(4) In low-power scenarios, if enabling control is required, apply R203, do not apply R200. Under no-load conditions, the maximum static power consumption of the DC-DC converter is 40uA, and when EN is turned off, the power consumption of the DC-DC converter is less than 1uA.
(5) If enabling control is not required, simply apply R200 without applying R203.

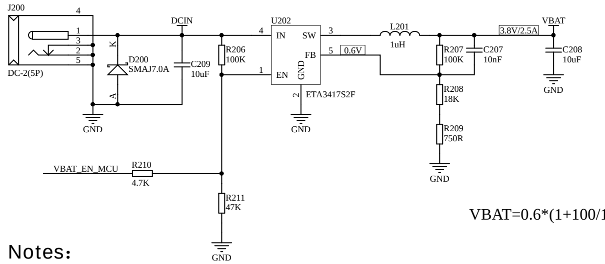
LDO（Power for Codec）



Notes:

(1) The LDO outputs a 3.3V voltage with a current capacity of 250mA, and it mainly supplies power to the Codec chip.
(2) This LDO has a static current of less than 1uA, making it highly suitable for low-power applications.

DC-DC BUCK

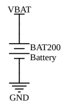


VBAT=0.6*(1+100/18.75)=3.80V

Notes:

(1) When the input power supply voltage differs significantly from the module's power supply voltage, choose DC-DC power supply.
(2) This circuit can handle a maximum input voltage of 7V and the output current can reach 2.5A.
(3) The selection of the D200 passive protection device should be based on the actual input voltage.
(4) The DC-DC circuit can be regarded as the pre-stage voltage reduction circuit, and then an LDO circuit can be connected after it to enhance the stability of the power supply.
(5) In low-power scenarios, if enabling control is required, apply R210, do not apply R206. Under no-load conditions, the maximum static power consumption of the DC-DC converter is 32uA, and when EN is turned off, the power consumption of the DC-DC converter is less than 1uA.
(6) If enabling control is not required, simply apply R206 without applying R210.

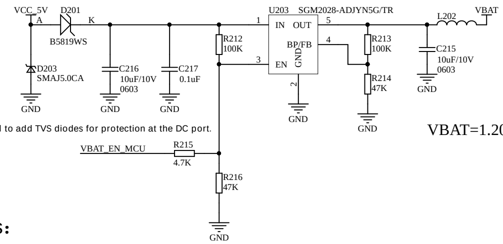
Battery



Notes:

(1) If battery power is chosen, the battery voltage should be between 3.3V and 4.5V. It can directly supply power.

LDO power supply scheme



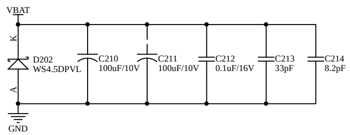
*It is recommended to add TVS diodes for protection at the DC port.

VBAT=1.206*(R213/R214+1)=3.77V

Notes:

(1) When the input power supply voltage is not significantly different from the power supply voltage of the module (for example, 5V), choose LDO power supply.
(2) This circuit can handle a maximum input voltage of 6V and the output current can reach 2A.
(3) The selection of the D203 passive protection device should be based on the actual input voltage.
(4) In low-power scenarios, if enabling control is required, apply R215, do not apply R212. Under no-load conditions, the maximum static power consumption of the LDO is 300uA, and when EN is turned off, the power consumption of the LDO is less than 1uA.
(5) If enabling control is not required, simply apply R212 without applying R215.

Module VBAT power supply



Notes:

(1) VBAT input voltage range: 2.2-4.5V.
(2) The capacitor should be placed close to the power pin of the module.

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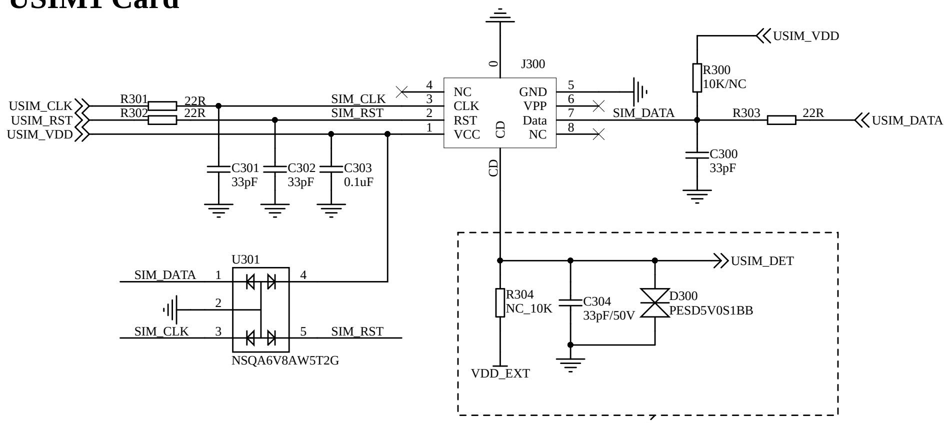
USIM1 Card

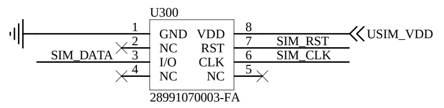


Hot-plug detection

Notes:

(1) The GND wiring of the SIM card should be short and thick. The decoupling capacitor of USIM_VDD should not exceed 1uF, and the capacitor should be placed close to the VCC of the external SIM card.
(2) Ground shielding should be added between the USIM_DATA, USIM_CLK and USIM_RST signals.If the USIM_DATA line is too long, it is recommended to reserve a 10K resistor externally and place it close to the card slot.
(3) The peripheral circuit of the SIM card should be placed close to the SIM card. The parasitic capacitance of the selected TVS diode should not exceed 15 pF, and it should be as close as possible to the external SIM card. Also, ensure that the SIM card signal line passes through the ESD protection device before reaching the module.
(4) A 22-ohm resistor needs to be connected in series between the module and the SIM card signal line to suppress stray EMI, and a 33pF capacitor should be connected in parallel to filter out RF interference.
(5) Hot-plug detection is only applicable to SIM card slots with detection pins. If the hot-plug detection function is not required, the corresponding circuitry for hot-plug detection can be omitted.

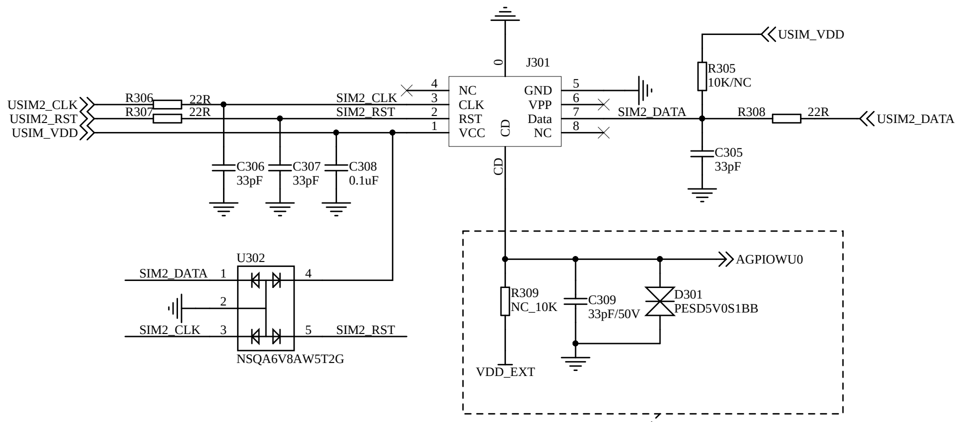
eSIM



Notes:

(1) If the eSIM card is close to the module and is not accessible to the user, the TVS diode can be omitted.

USIM2 Card



Hot-plug detection

备注:

(1) Here, the AGPIOWU0 pin is used for hot-plug detection, which requires software support. Alternatively, other pins can also be selected for the detection.
(2) If the hot-plug detection function is not required, the hot-plug detection circuitry can be removed. At the same time, the hot-plug detection function of the module can be disabled by coordinating with the corresponding AT instructions.

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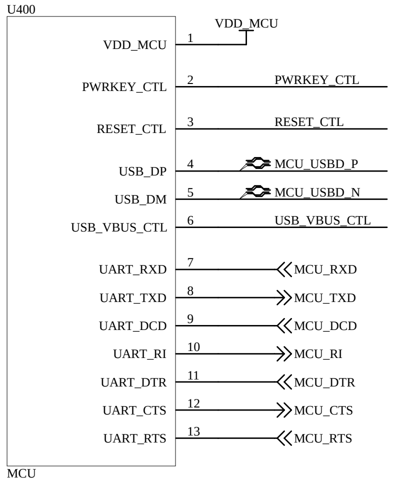
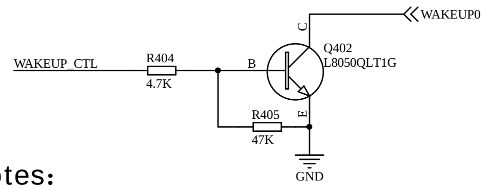
MCU interface



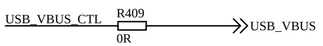
WAKEUP circuit



Notes:

(1) When using the MCU (IO:WAKEUP_CTL) to control the power-on/off pins of the module, it is preferable to use an open-collector drive circuit for control.
(2) It is recommended to reserve a 0.1uF capacitor at the module pin position. By default, do not apply it.
(3) The WAKEUP1 control circuit can be referred to the above circuit.

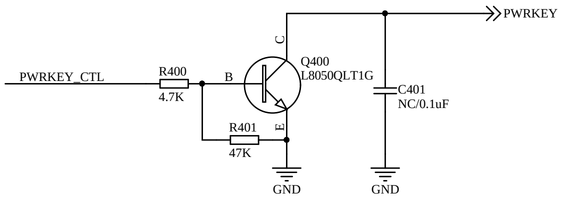
USB_VBUS circuit

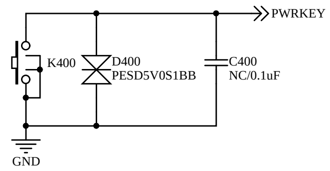


Notes:

(1) This circuit can also wake up the module by controlling the USB_VBUS of the module through the USB_VBUS_CTL of the MCU.
(2) If it is an external USB interface, USB_5V connects to USB_VBUS, and by inserting the USB cable, the module can be awakened.

Power on/off reference circuit (controlled by MCU or key press, either option is acceptable)





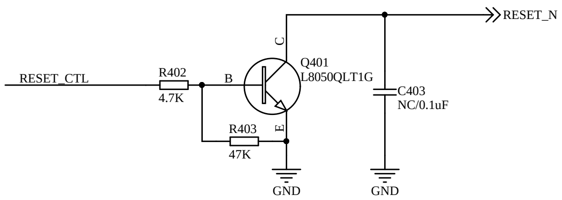
Notes:

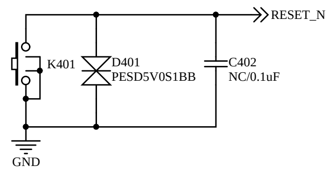
(1) When using the MCU (IO:PWRKEY_CTL) to control the power-on/off pins of the module, it is preferable to use an open-collector drive circuit for control.
(2) It is recommended to reserve a 0.1uF capacitor at the module pin position. By default, d
not apply it.

Notes:

(1) When using the MCU (IO:PWRKEY_CTL) to control the power-on/off pins of the module, it is preferable to use an open-collector drive circuit for control.
(2) To achieve better electrostatic protection performance, it is recommended to place the ESD components close to the button.

Reset reference circuit (controlled by MCU or key press, either option is acceptable)





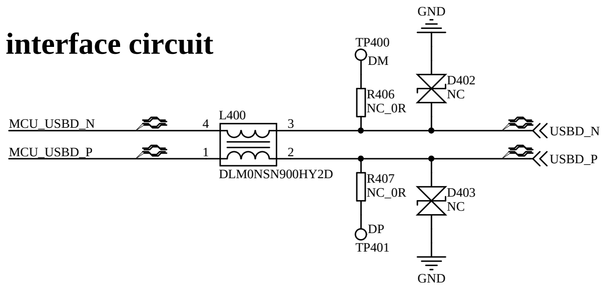
Notes:

(1) When using the MCU (IO:RESET_CTL) to control the power-on/off pins of the module, it is preferable to use an open-collector drive circuit for control.
(2) It is recommended to reserve a 0.1uF capacitor at the module pin position. By default, d
not apply it.

Notes:

(1) When using the MCU (IO:PWRKEY_CTL) to control the power-on/off pins of the module, it is preferable to use an open-collector drive circuit for control.
(2) To achieve better electrostatic protection performance, it is recommended to place the ESD components close to the button.

USB interface circuit



Notes:

(1) It is recommended to add a common-mode inductor to the USB cable between the MCU and the module to reduce EMI interference.
(2) Common-mode inductors, resistors and test points are placed close to the module.
(3) USB cable control differential 90-ohm impedance.

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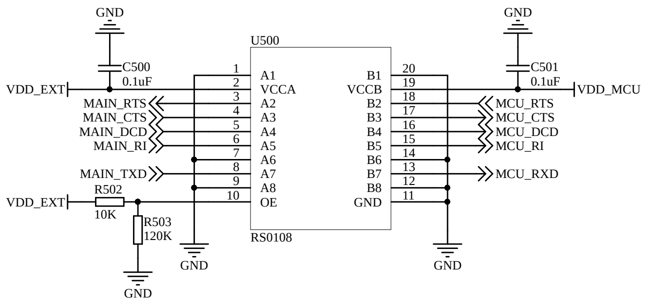
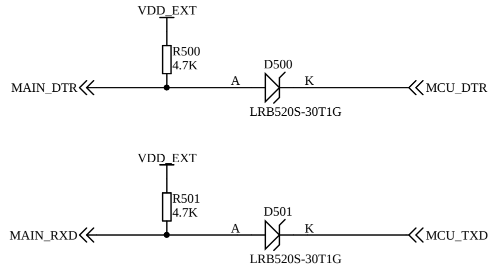
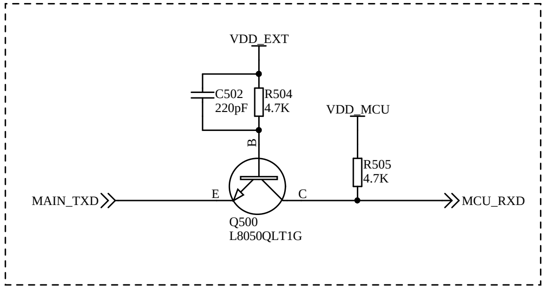
Level switching circuit



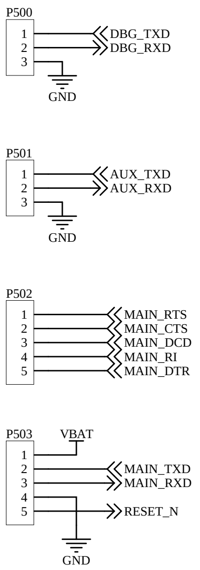




Notes:

(1) When using the level conversion chip circuit, it is necessary to note that VCCA ≤ VCCB. For specific details, please refer to the chip datasheet.
(2) The chip enable pin OE is connected to the reference voltage VCCA. By pulling up the OE pin, a two-way level conversion is achieved.
(3) Level conversion chip rate: Push-pull mode 20 Mbps, Open-collector mode 2 Mbps.
(4) Since VDD_EXT is not powered during sleep mode, the chip-based solution cannot be used for MAIN_RXD and MAIN_DTR.
(5) Using the discrete component solution, the MAIN_RTS, MAIN_DTR conversion circuits are similar to the MAIN_RXD circuit.
(6) Using the discrete component solution, the MAIN_CTS, MAIN_RI, and MAIN_DCD conversion circuits are similar to the MAIN_TXD circuit.
(7) When using the diode conversion circuit, a low-dropout (≤0.3V) Schottky diode should be selected.

UART interface



Notes:

(1) DBG_UART can be used to view the LOG data of the module. It is recommended to reserve test points.
(2) AUX_UART can be used for the custom serial port function of the customer, but the function is not yet supported.
(3) MAIN_UART can be used for data transmission, AT commands and firmware upgrade.It supports hardware flow control and low-power wake-up. It is recommended to reserve test points.
(4) When connecting the MCU to the module UART, be sure to pay attention to the issue of level matching.
(5) MAIN_RI and MAIN_DTR can maintain the voltage output even in the sleep mode.

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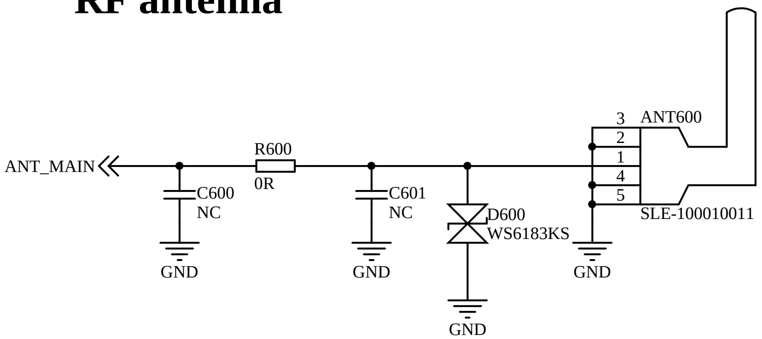
RF antenna



Notes:

(1) It is recommended to reserve a π-type antenna matching circuit. The actual value will depend on the antenna.
(2) To achieve better electrostatic protection performance, it is recommended to place the ESD components close to the antenna.
(3) The parasitic capacitance of the selected ESD diode should not exceed 0.5 pF.

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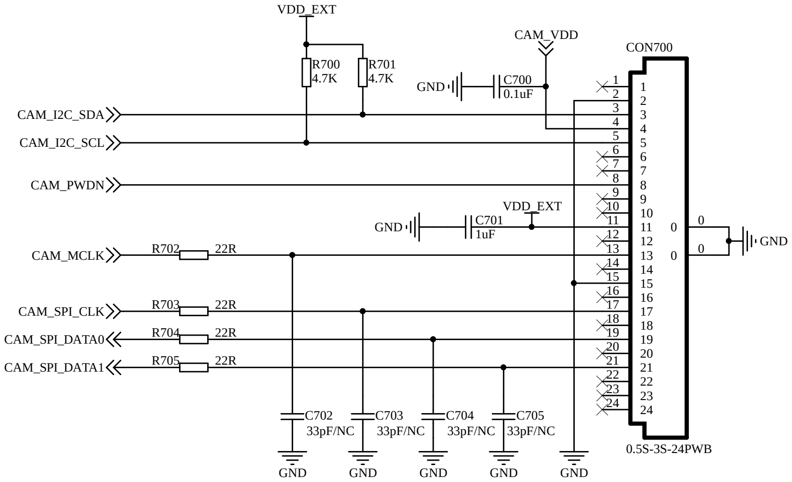
CAMERA

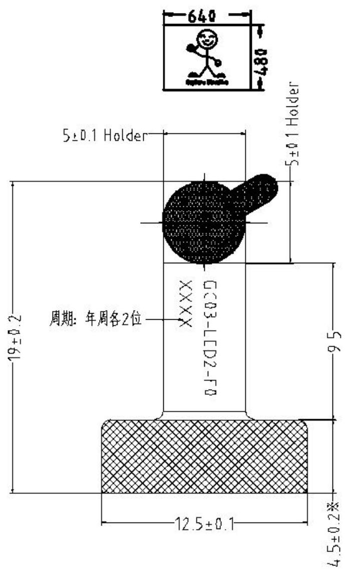


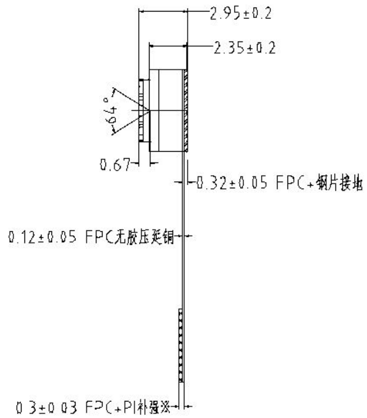
1	NC
2	GND
3	SDA
4	AVDD2.8V
5	SCL
6	NC
7	NC
8	PWDN
9	NC
10	NC
11	DOVDD
12	NC
13	MCLK
14	NC
15	GND
16	NC
17	PCLK
18	NC
19	DO
20	NC
21	DI
22	NC
23	NC
24	NC

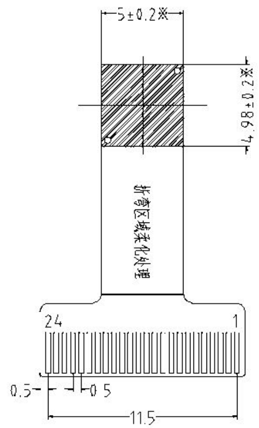
Notes：

(1) This reference design uses the GC03-LED2-F0 camera. Cameras of other models have similar principles, but some details may need to be modified.
(2) In this reference design, the camera is connected to the module through the FPC socket interface. It is important to note that the wiring sequence of the FPC socket should correspond to the wiring sequence of the camera's pins.
(3) CAM_VDD is provided by the module, with a voltage of 2.8V and a current capacity of 300mA. It is recommended to disable the enable of CAM_VDD when not in use to reduce the power consumption of the module.
(4) Resistors R702 to R705 should be placed close to the signal source end. The resistance value can be adjusted according to the signal quality. The recommended resistance value is 22Ω to 68Ω.
(5) Capacitors C702 to C705 can be left unattached. During actual application, the capacitance value can be adjusted according to the quality of the signal. It is recommended to use a capacitance value ranging from 22pF to 68pF.

Camera model：GC03-LED2-F0
chip type：GC032A







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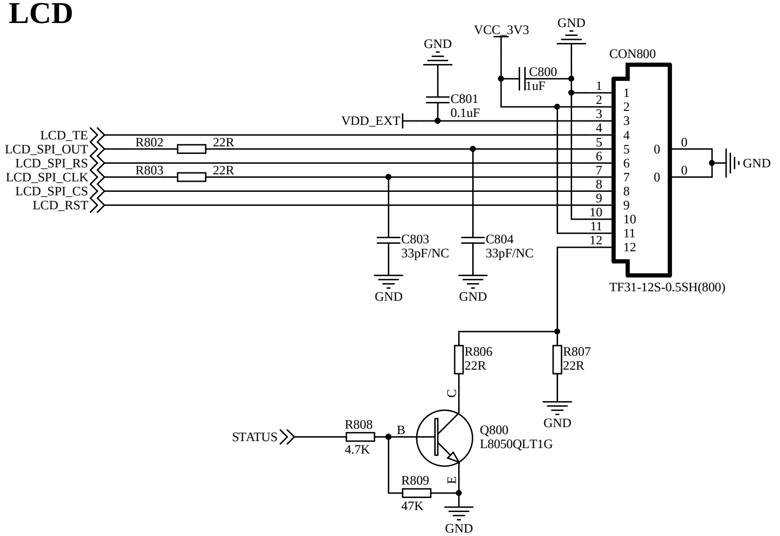
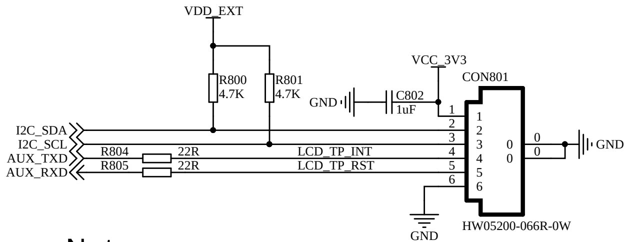
LCD



1	GND
2	VCC
3	IOVCC
4	TE
5	SDA
6	DC
7	SCL
8	CS
9	RST
10	GND
11	A
12	K



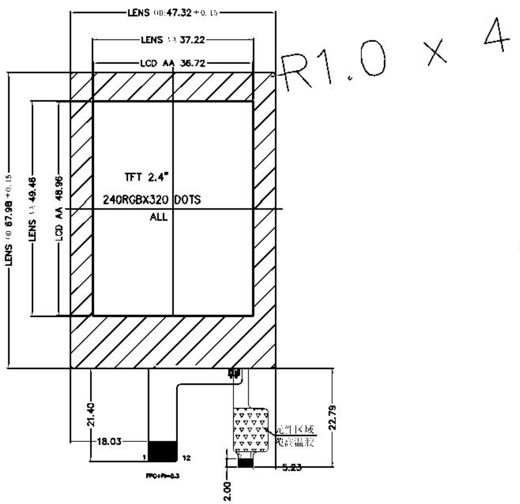
1	VDD
2	SDA
3	SCL
4	INT
5	RST
6	GND

Notes:

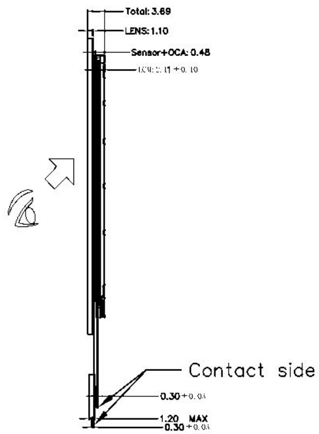
(1) The LCD screen size parameters selected in this plan are as follows. If other LCD screens are to be used, some configuration or modification may be required in the hardware or software.
(2) The module does not provide dedicated pins for powering the LCD screen. An independent LDO power supply is required to output 3.3V to power the LCD screen.
(3) If you want to use the touch function of the LCD screen, you will need an additional I2C line and two GPIOs. This reference design uses the AUX UART to re-purpose the touch screen control pins of the LCD.
(4) The LCD screen of this scheme is connected to the module pins through an FPC socket. When designing, it is necessary to ensure that the wiring sequence of the FPC socket and the FPC pins is consistent.
(5) If the LCD backlight needs to remain constantly on, solder R807; remove R807 and solder R806, R808, R809 and Q800 instead. Then the LCD backlight can be controlled through IO. This reference design uses STATUS for control.
(6) The R802 and R803 resistors should be placed close to the signal source end. The resistance value can be adjusted according to the signal quality. The recommended resistance value is 22Ω to 68Ω. The R804 and R805 positions are reserved for easy debugging.
(7) Capacitors C803 and C804 can be left unattached. During actual application, the capacitance value can be adjusted according to the quality of the signal. It is recommended to use a capacitance value ranging from 22pF to 68pF.

LCD chip model：ST7789V3
Touchscreen chip model：FT6206

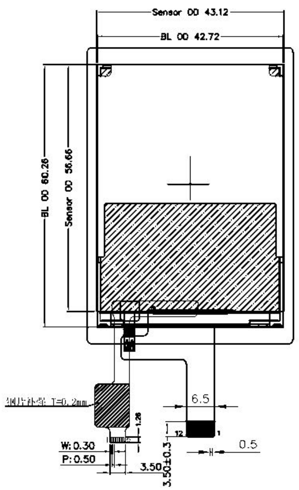
Front



Side



Rear



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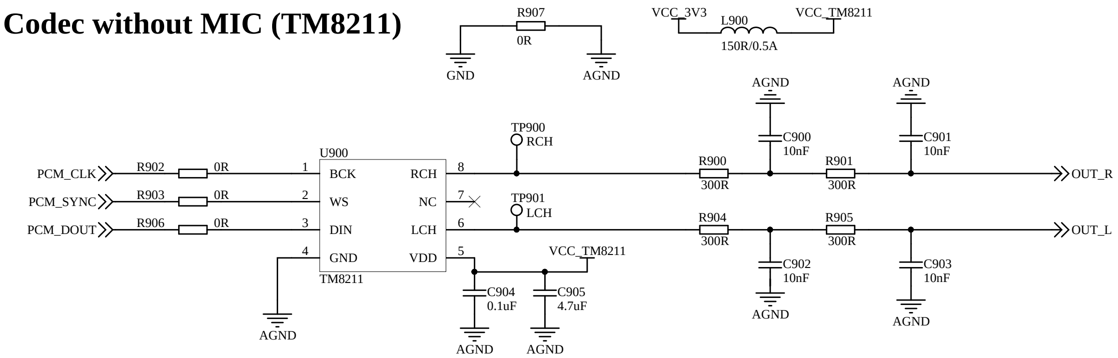
This reference design offers three options for Codec chips. Users can choose the one that suits their needs.

Codec without MIC (TM8211)



Notes:

(1) The analog output of the audio decoder can only drive low-power loads such as headphones. For other high-power loads like speakers, an audio power amplifier needs to be added during the design process.
(2) Both the OUT_R and OUT_L output signal lines need to be surrounded by GND nets to keep them away from interference sources.
(3) When the circuit design has a high requirement for low-noise output, it is recommended that the audio circuit be powered by an independent power supply.
(4) Pay attention to the distinction between analog and digital grounds. They can be connected through an 0R resistor.
(5) The recommended operating voltage for TM8211 is 3 - 8V, with a typical value of 5V. Here, 3.3V is used as the power supply.

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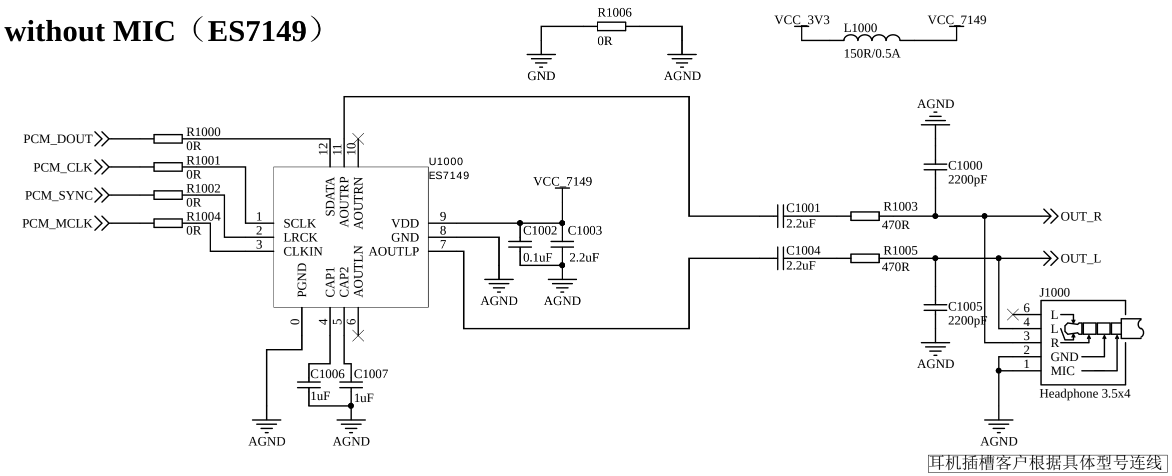
Codec without MIC（ES7149）



Notes:

(1) The analog output of the audio decoder can only drive low-power loads such as headphones. For other high-power loads like speakers, an audio power amplifier needs to be added during the design process.
(2) Both the OUT_R and OUT_L output signal lines need to be surrounded by GND nets to keep them away from interference sources.
(3) When the circuit design has a high requirement for low-noise output, it is recommended that the audio circuit be powered by an independent power supply.
(4) Pay attention to the distinction between analog and digital grounds. They can be connected through an 0R resistor.
(5) The recommended operating voltage for ES7149 is 3 - 3.6V, with a typical value of 3.3V. Here, 3.3V is used as the power supply.
(6) ES7149 also supports differential output. The 6th and 7th pins are a pair of differential outputs for the left channel, while the 10th and 11th

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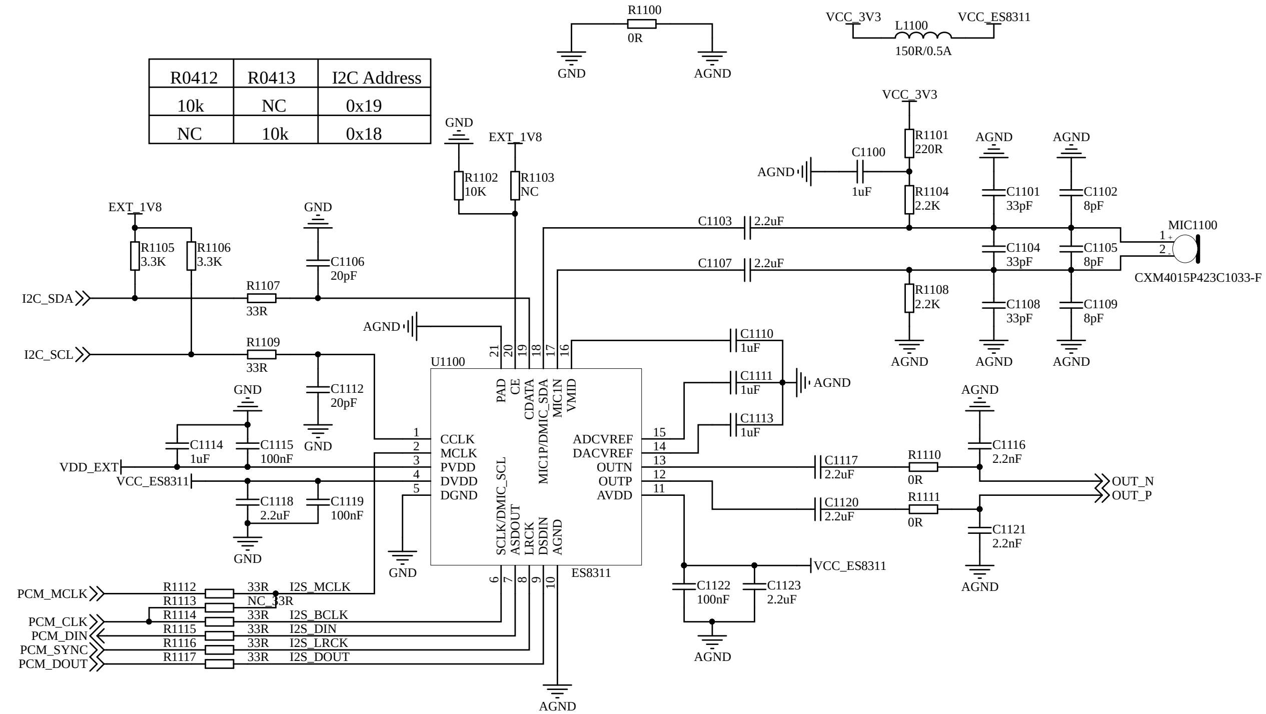
Codec with MIC（ES8311）

R0412	R0413	I2C Address
10k	NC	0x19
NC	10k	0x18



Notes:

(1) The analog output of the audio decoder can only drive low-power loads such as headphones. For other high-power loads like speakers, an audio power amplifier needs to be added during the design process.
(2) Both the OUT_R and OUT_L output signal lines need to be surrounded by GND nets to keep them away from interference sources.
(3) When the circuit design has a high requirement for low-noise output, it is recommended that the audio circuit be powered by an independent power supply.
(4) Pay attention to the distinction between analog and digital grounds. They can be connected through an 0R resistor.
(5) For ES8311, the recommended operating voltage for the analog power supply AVDD is 1.7V to 3.6V, with a typical value of 3.3V; the recommended operating voltage for the PVDD and DVDD power supplies is 1.6V to 3.6V, with a typical value of 3.3V; the PVDD voltage should be consistent with the voltage of PCM and other digital signals.

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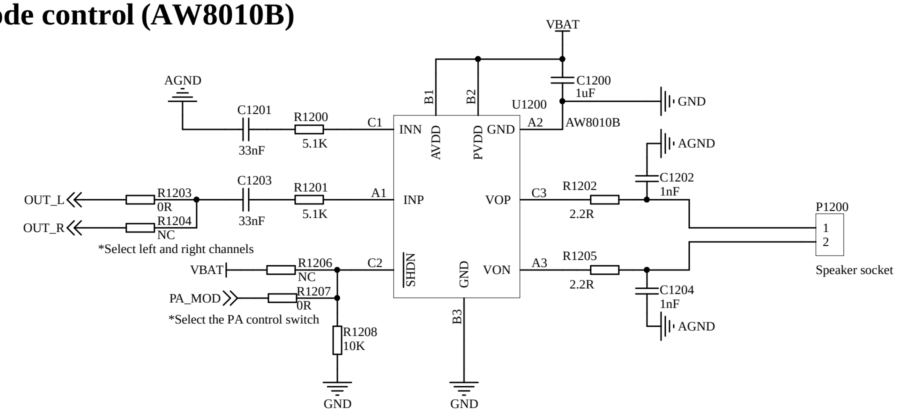
This reference design offers two options for audio PA chips. Users can choose the one that suits their needs.

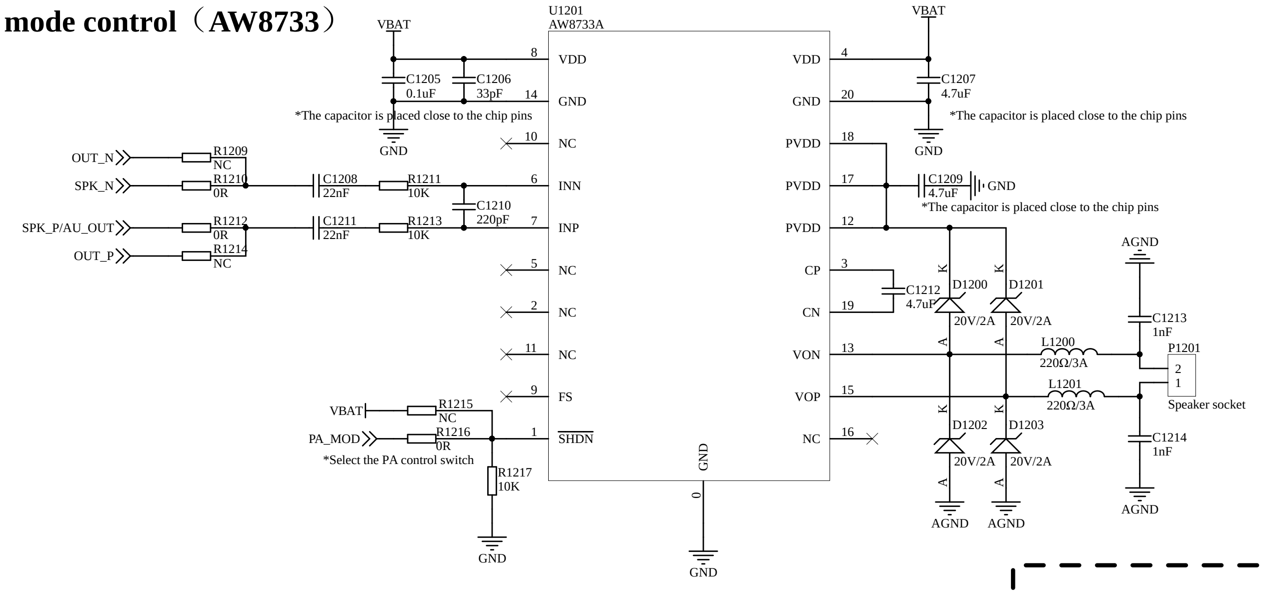
PA without mode control (AW8010B)



Notes:

(1) The recommended operating voltage for AW8010B is 2.5 - 5.5V. Here, connect to VABT.
(2) The magnification can be adjusted by R1200 and R1201. The calculation method is 315KΩ/Rin. For more details, please refer to the device specification sheet.
(3) The SHDN pin is the shutdown pin of PA. turning it off with a low voltage. If there is no need for shutdown, it can be directly connected to VBAT; if it is necessary to control the shutdown, a GPIO can be connected instead.
(4) The resistors and capacitors at the output end should be placed close to the VON and VOP pins of the chip. The output line from the chip to the speaker should be as short and thick as possible.

PA with mode control（AW8733）



Notes:

(1) When laying out the PCB, PIN4 and PIN8 (VDD) must be connected together and all should be connected to VABT.
(2) When laying out the PCB, PIN12, PIN17, and PIN18 (PVDD) must be connected together.
(3) The input capacitors and input resistors should be placed as close as possible to the INN and INP pins of the chip, and the input lines should run parallel to each other to suppress noise coupling.
(4) The magnetic beads and capacitors should be placed close to the VON and VOP pins of the chip. The output wires from the chip to the speaker should be as short and thick as possible.
(5) The SHDN pin is the control pin for PA. It can control the switch of PA, turning it off with a low voltage; it can also control the gain of PA, as detailed in the device specification.
If there is no need to control the switch and gain of PA, it can be directly connected to VBAT. If control is required, a GPIO can be connected ir

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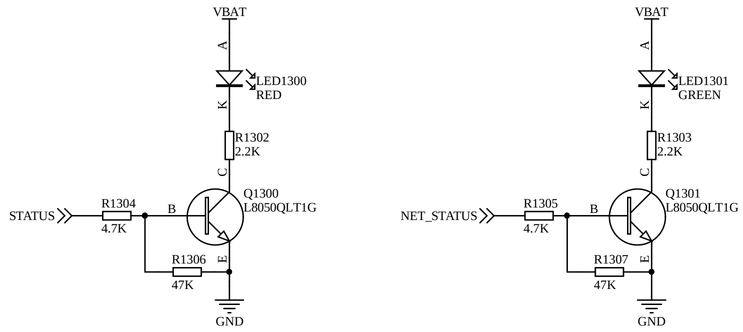
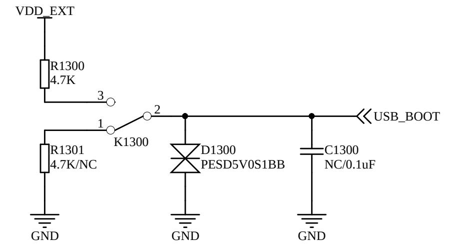
STATUS & NET_STATUS Reference Circuit



Notes:

(1) STATUS is the module status indicator pin. After the module is worked on normally, STATUS outputs a high level.
(2) NET_STATUS is the network status indicator pin, which will adjust the duration of high and low voltage levels according to the network status of the module.

USB_BOOT Reference Circuit



Notes:

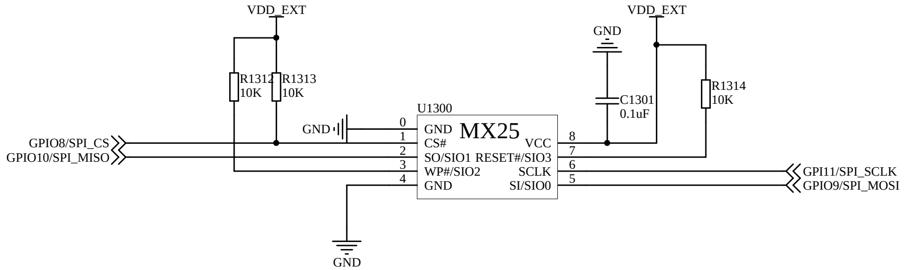
(1) Before powering on, pulling up USB_BOOT can enable the module to enter the emergency download mode after startup, and then the firmware can be downloaded via the USB interface.
(2) USB_BOOT should remain disconnected by default.

GPIO

Notes:

(1) GPIO12 and GPIO13 are general-purpose GPIOs, with the reference voltage level being VDD_EXT.
(2) AGPIO3 and AGPIO5 are AONGPIO. The sleep mode can be configured without power loss.
(3) If not used, it is recommended to leave it hanging.

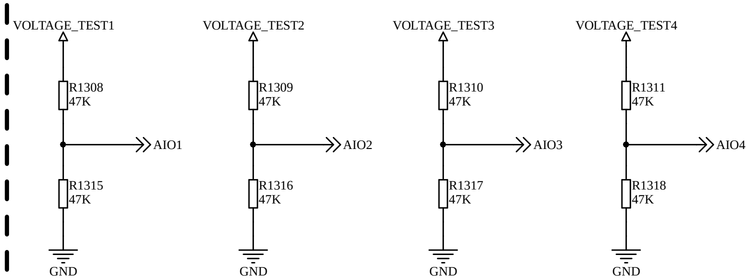
FLASH



Notes:

(1) The module only supports the standard SPI interface for external connection of FLASH.

ADC



Notes:

(1) VOLTAGE_TEST indicates the voltage to be tested.
(2) The input voltage range for AIO1 to AIO4 is 0-1.5V.

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3 Related documents and terminology abbreviations

The following documents provide the names of the relevant documents, please refer to the latest release for specific versions.

Table 3-1 Related Documents

Serial Number	Document Name	Comment
[1]	Lierda NT26-FCN B Series Hardware Design Manual	

Table 3-2 Abbreviations

Abbreviation	Full English name	Chinese full name
ANT	Antenna	Antenna
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC converter
GPIO	General-purpose input/output	General Input Output
LDO	Low Dropout Regulator	Low Dropout Linear Regulator
MCU	Mirco Controller Unit	Microcontroller
NB-IoT	Narrow Band Internet of Things	Narrowband Internet of Things
PSM	Power Saving Mode	Energy saving mode
RF	Radio Frequency	Radio frequency
RX	Receive	Receive
TVS	Transient Voltage Suppressor	Transient Voltage Suppression Diode
TX	Transmit	Send
UART	Universal Asynchronous Receiver & Transmitter	General asynchronous receiver and transmitter
USIM	Universal Subscriber Identification Module	Universal User Identification Module